Vector Differentiation

★ Ordinary Differentiation of Scalar Function

Let $\phi = \phi(u)$ be the scalar point function define for any variable u. then

$$\frac{d\phi}{du} = \lim_{\delta u \to 0} \frac{\delta \phi}{\delta t} = \lim_{\delta u \to 0} \frac{\phi(u+\delta u) - \phi(u)}{\delta u}$$

is known as derivative of scalar point function $\phi(u)$.

★ Ordinary Derivative of Vectors

Let $R(u)$ be a vector depending on scalar u.

Then

$$\lim_{\delta u \to 0} \frac{\delta R}{\delta u} = \frac{R(u+\delta u) - R(u)}{\delta u} = \frac{d\,R(u)}{du}$$

is known as ordinary derivative of $R(u)$ with respect to u (provided limit should exist.)

Since $\frac{d\,R(u)}{du}$ it self a vector depending on u

we can consider it's derivative w.r.to u.

If derivative is exist then it is denoted by $\frac{d^2R}{du^2}$

in the similar manner we can discribe higher derivative of R w.r.to. u.

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Space curve

Let (x, y, z) be the any point in the space (3-Dimensional space). and

$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ represent the position vector of pt (x, y, z) from the origin.

If x, y & z are parametric curve of scalar u

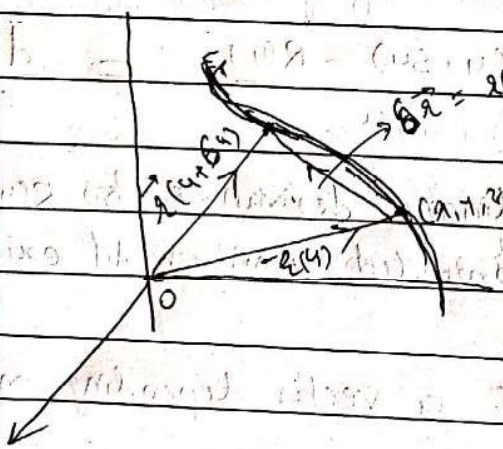
i.e. $x = x(u)$, $y = y(u)$, & $z = z(u)$

Then

$$\frac{d\vec{r}}{du} = \frac{d}{du}\left[x(u)\,\hat{i} + y(u)\,\hat{j} + z(u)\,\hat{k}\right]$$

$$= \left(\frac{dx}{du}\right)\hat{i} + \left(\frac{dy}{du}\right)\hat{j} + \left(\frac{dz}{du}\right)\hat{k}$$

Represent derivative of $\vec{r}$ w.r.to u.



If u is the time 't'

Then

$\frac{d\vec{r}}{dt}$ represent a velocity $\vec{v}$

Similarly $\frac{d^2\vec{r}}{dt^2} = \frac{d\vec{v}}{dt} = \vec{a}$ represents acceln along the curve.

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ie

$$\vec{a} = \text{acc}^n = \frac{d^2\vec{r}}{dt^2}$$

$$= \frac{d}{dt}\left(\frac{d\vec{r}}{dt}\right)$$

$$= \frac{d}{dt}\left[\frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k}\right]$$

$$\vec{a} = \frac{d^2x}{dt^2}\hat{i} + \frac{d^2y}{dt^2}\hat{j} + \frac{d^2z}{dt^2}\hat{k}$$

Continuity and Differentiability

let $\phi = \phi(u)$ be the scalar function. it is said to be continuous at some u if

$$\lim_{\delta u \to 0} \phi(u+\delta u) = \phi(u)$$

OR

A function $\phi(u)$ is said to be continuous at u if for some positive number $\in$, there exist a some positive number δ such that

when $|\delta u| < \delta \Rightarrow |\phi(u+\delta u) - \phi(u)| < \in$

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Now: let $R(u) = R_1(u)\,\hat{i} + R_2(u)\,\hat{j} + R_3(u)\,\hat{k}$ be the vector point function it is said to be continuous at u if

$$\lim_{\delta u \to 0} \vec{R}(u+\delta u) = \vec{R}(u) \qquad \text{(easy)}$$

OR

A vector $\vec{R}(u) = R_1\,\hat{i} + R_2\,\hat{j} + R_3\,\hat{k}$ is said to be continuous at u if for some positive number ϵ, there exist some δ positive number δ. such that

$$\text{for } |\delta u| < \delta \Rightarrow \left|\vec{R}(u+\delta u) - \vec{R}(u)\right| < \epsilon$$

A function (scalar or vector) is said to be differentiable of nth order its its nth derivative is exist.

(~~But~~ If a function (scalar or vector) is differentiable then it is obvious continuous. but the converse is ~~not~~ need not to be true.)

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If $\bar{f}(t) = f_1(t)\,\hat{i} + f_2(t)\,\hat{j} + f_3(t)\,\hat{k}$ be a vector function of t

where f_1, f_2, f_3 are scalar functions.

then

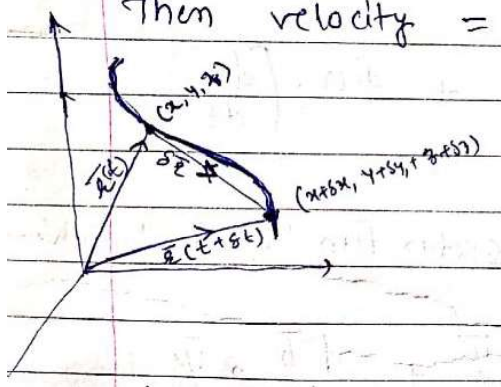
$$\boxed{\frac{d\bar{f}(t)}{dt} = \left(\frac{df_1}{dt}\right)\bar{i} + \left(\frac{df_2}{dt}\right)\bar{j} + \left(\frac{df_3}{dt}\right)\bar{k}}$$

Velocity and Acceleration

Consider a particle moving along a curve in space at time t, the position vector is defined as (at pt (x,y,z))

$$\bar{r} = x(t)\,\hat{i} + y(t)\,\hat{j} + z(t)\,\hat{k}$$

Then velocity = Rate of change of displacement at time t



$$\boxed{V = \frac{d\bar{r}}{dt} = \frac{dx(t)}{dt}\hat{i} + \frac{dy(t)}{dt}\hat{j} + \frac{dz(t)}{dt}\hat{k}}$$

it is also denoted as $\dot{\bar{r}}$

Now acel^n = Rate of change of velocity w.r.t time t.

$$\therefore \bar{a} = \frac{d\bar{v}}{dt} = \left(\frac{d^2x}{dt^2}\right)i + \left(\frac{d^2y}{dt^2}\right)j + \left(\frac{d^2z}{dt^2}\right)k$$

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Some Standard Results

(1) if $\bar{k}$ is a constant vector, then

$$\frac{d\bar{k}}{dt} = 0,$$

(2) If $\phi(t)$, $\psi(t)$ and $\xi(t)$ are ~~vector~~ differentiable vector function of a scalar variable t. then

- $\frac{d}{dt}(k\phi(t)) = k \cdot \frac{d\phi}{dt}$ where k is any scalar

- $\frac{d}{dt}(\phi(t) \pm \psi(t)) = \frac{d\phi}{dt} \pm \frac{d\psi}{dt}$

- $\frac{d}{dt}(\bar{a} \cdot \bar{\phi}(t)) = \bar{a} \cdot \left(\frac{d\bar{\phi}(t)}{dt}\right) + \bar{\phi}(t) \cdot \left(\frac{d\bar{a}}{dt}\right)$ $= 0$

where $\bar{a}$ is scalar function of variable t.

- $\frac{d}{dt}(\bar{\phi}(t) \cdot \bar{\psi}(t)) = \left(\bar{\phi} \cdot \frac{d\bar{\psi}}{dt}\right) + \left(\frac{d\bar{\phi}}{dt} \cdot \bar{\psi}\right)$

- $\frac{d}{dt}(\bar{\phi}(t) \times \bar{\psi}(t)) = \left(\bar{\phi} \times \frac{d\bar{\psi}}{dt}\right) + \left(\frac{d\bar{\phi}}{dt} \times \bar{\psi}\right)$

- $\frac{d}{dt}[\bar{\phi}(t)\ \bar{\psi}(t)\ \bar{\xi}(t)] = \left[\frac{d\bar{\phi}}{dt}, \bar{\psi}(t), \bar{\xi}(t)\right]$

$$+ \left[\phi(t), \frac{d\bar{\psi}}{dt}, \xi(t)\right] + \left[\phi, \psi, \frac{d\bar{\xi}}{dt}\right]$$

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ie.

$$\frac{d}{dt}\left\{\bar{\phi}\circ(\bar{\psi}\times\bar{\xi})\right\} = \bar{\phi}\circ\frac{d}{dt}(\bar{\psi}\times\bar{\xi}) + \frac{d\bar{\phi}}{dt}\circ(\bar{\psi}\times\bar{\xi})$$

$$= \bar{\phi}\circ\left\{\bar{\psi}\times\frac{d\bar{\xi}}{dt} + \frac{d\bar{\psi}}{dt}\times\bar{\xi}\right\} + \frac{d\bar{\phi}}{dt}\cdot(\bar{\psi}\times\bar{\xi})$$

$$= \bar{\phi}\circ\left(\bar{\psi}\times\frac{d\bar{\xi}}{dt}\right) + \bar{\phi}\circ\left(\frac{d\bar{\psi}}{dt}\times\bar{\xi}\right) + \frac{d\bar{\phi}}{dt}\circ(\bar{\psi}\times\bar{\xi})$$

$$= \left[\bar{\phi},\bar{\psi},\frac{d\bar{\xi}}{dt}\right] + \left[\bar{\phi},\frac{d\bar{\psi}}{dt},\bar{\xi}\right] + \left[\frac{d\bar{\phi}}{dt},\bar{\psi},\bar{\xi}\right]$$

- $\frac{d}{dt}\left[\phi\times(\psi\times\xi)\right]$

$$= \bar{\phi}\times\left(\psi\times\frac{d\bar{\xi}}{dt}\right) + \bar{\phi}\times\left(\frac{d\psi}{dt}\times\bar{\xi}\right) + \frac{d\bar{\phi}}{dt}\times(\psi\times\xi)$$

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Tangent vector to a curve at point $P(x(t), y(t), z(t))$

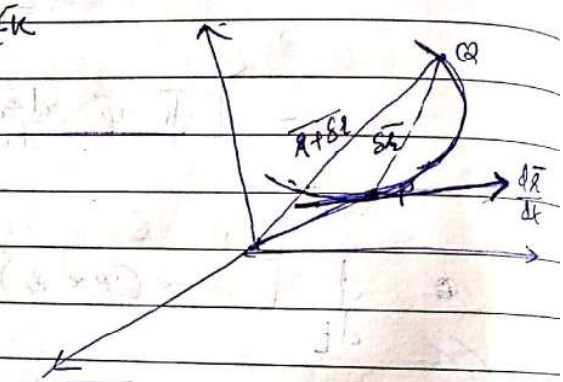
let $f(t)$ be a space curve and $P(x,y,z)$.
& $Q(x+\delta x, y+\delta y, z+\delta z)$ be any two point on a curve

where $x = x(t)$, $y = y(t)$, $z = z(t)$. & $\bar{r} = x(t)\,i + y(t)\,j + z(t)$ be the position vector

~~Then tangent at pt. P. is given by.~~

Then tangent at point P. is given by

$$\frac{d\bar{r}}{dt} = \frac{dx}{dt}\,i + \frac{dy}{dt}\,j + \frac{dz}{dt}\,k$$



Tangential component of a vector $\bar{a}$

let $\bar{a}$ be the any vector, then it's tangential component is denoted and defined as

$$a_t = \bar{a} \cdot \hat{t}$$

where $\hat{t}$ is unit tangent vector $= \dfrac{d\bar{r}/dt}{|d\bar{r}/dt|}$

① **Note:** If $\bar{a}$ is accln.
Then tangential component of accln is the measure of rate of change in the magnitude of the velocity vector (i.e. speed)]

② and Normal component of accln is the measure of rate of change of the direction of velocity vector]

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Normal component of vector $\bar{a}$

It is denoted and defined as

$$\boxed{a_n = |\hat{t} \times \bar{a}|} = \text{magnitude of } (\hat{t} \times \bar{a})$$

where $\hat{t}$ is unit tangent vector.

Ex :⇒ let $\bar{a} = t^2\hat{i} - t\hat{j} + (2t+1)\hat{k}$, $\bar{b} = (2t-3)\hat{i} + \hat{j} - t\hat{k}$

Find (i) $\frac{d}{dt}(\bar{a}\cdot\bar{b})$ (ii) $\frac{d}{dt}(\bar{a}\times\bar{b})$ (iii) $\frac{d}{dt}(\bar{a}\times\frac{d\bar{b}}{dt})$.

at ~~time~~ $t = 1$

Soln: Given $\bar{a} = t^2\hat{i} - t\hat{j} + (2t+1)\hat{k}$

$\bar{b} = (2t-3)\hat{i} + \hat{j} - t\hat{k}$

$\therefore \bar{a}\cdot\bar{b} = [t^2\hat{i} - t\hat{j} + (2t+1)\hat{k}] \cdot [(2t-3)\hat{i} + \hat{j} - t\hat{k}]$

$= t^2(2t-3) + (-t)(1) + (2t+1)(-t)$

$\boxed{\bar{a}\cdot\bar{b} = 2t^3 - 5t^2 - 2t}$

$\therefore \hat{i}\cdot\hat{i} = 1$
$\hat{j}\cdot\hat{j} = 1$
$\hat{k}\cdot\hat{k} = 1$
$\hat{i}\cdot\hat{k} = 0$
$\hat{j}\cdot\hat{k} = 0$
$\hat{k}\cdot\hat{i} = 0$

$\therefore \frac{d}{dt}(\bar{a}\cdot\bar{b}) = \frac{d}{dt}(2t^3 - 5t^2 - 2t)$

$= 6t^2 - 10t - 2$

at $t = 1$

$\frac{d(\bar{a}\cdot\bar{b})}{dt} = 6 - 10 - 2 = \boxed{-6}$

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(ii) Now

$$\bar{a}\times\bar{b} = \begin{vmatrix} i & j & k \\ t^2 & -t & (2t+1) \\ (2t+3) & 1 & -t \end{vmatrix}$$

$$= i\left[t^2-(2t+1)\right] - j\left[-t^3-(2t+1)(2t-3)\right] + k\left[t^2+t(2t-3)\right]$$

$$\bar{a}\times\bar{b} = (t^2-2t-1)\hat{i} + (3t^3+4t^2-4t-3)\hat{j} + (3t^2-3t)\hat{k}$$

$$\frac{d}{dt}(\bar{a}\times\bar{b}) = \frac{d}{dt}\left[(t^2-2t+1)\hat{i} + (3t^3+4t^2-4t-3)\hat{j} + (6t-3)\hat{k}\right]$$

$$= (2t-2)\hat{i} + (3t^2+8t-4)\hat{j} + 6\hat{k}$$

$\therefore$ at $t=1$

$$\frac{d}{dt}(\bar{a}\times\bar{b}) = 0\hat{i} + 7\hat{j} + 6\hat{k} = 7\hat{j}+3\hat{k}$$

(iii) let $\bar{b} = (2t-3)\hat{i} + \hat{j} - t\hat{k}$

$$\therefore \frac{d\bar{b}}{dt} = 2\hat{i} + 0\hat{j} - \hat{k}$$

Thus

$$\bar{a}\times\frac{d\bar{b}}{dt} = \begin{vmatrix} i & j & k \\ t^2 & -t & (2t+1) \\ 2 & 0 & -1 \end{vmatrix}$$

$$= t\hat{i} + (t^2+4t-2)\hat{j} + 2t\hat{k}$$

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$$\therefore \frac{d}{dt}\left(\bar{a} \times \frac{d\bar{b}}{dt}\right) = \frac{d}{dt}\left[t\hat{i} + (t^2+4t-2)\hat{j} + 2t\hat{k}\right]$$

$$= \hat{i} + (2t+4)\hat{j} + 2\hat{k}$$

at $t=1$

$$\frac{d}{dt}\left(\bar{a} \times \frac{d\bar{b}}{dt}\right) = \hat{i} + 6\hat{j} + 2\hat{k}$$

Ex: Find the unit tangent vector at point on the curve given by $x = (t^2+2)$, $y = (4t-5)$, $z = (2t^2-6t)$ at $t=2$

where t is any variable ~~at such point~~

Soln: let $x(t) = t^2+2$, $y(t) = 4t-5$, $z(t) = 2t^2-6t$

let $\bar{r} = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$ be the position vector of any point (x, y, z) on the curve.

i.e. $\bar{r} = (t^2+2)\hat{i} + (4t-5)\hat{j} + (2t^2-6t)\hat{k}$

$\therefore$ Tangent at pt (x, y, z) is given by

$$\frac{d\bar{r}}{dt} = \frac{d}{dt}\left[(t^2+2)\hat{i} + (4t-5)\hat{j} + (2t^2-6t)\hat{k}\right]$$

$$= 2t\hat{i} + 4\hat{j} + (4t-6)\hat{k}$$

$\therefore$ at ~~point~~ $t=2$

$$\frac{d\bar{r}}{dt} = 4\hat{i} + 4\hat{j} + 2\hat{k}$$

$$\therefore \left|\frac{d\bar{r}}{dt}\right| = \sqrt{(4)^2 + (4)^2 + (2^2)} = \sqrt{36} = 6$$

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∴ Unit tangent vector ~~at t=0~~

$$\hat{t} = \frac{d\bar{r}/dt}{|d\bar{r}/dt|}$$

$$\hat{t} = \frac{1}{6}(4\hat{i} + 4\hat{j} + 2\hat{k})$$

$$\boxed{\hat{t} = \frac{1}{3}(2\hat{i} + 2\hat{j} + \hat{k})}$$

Ex: Find the tangential and Normal component of acc^n at any time t. of a particle whose position at time t is given by

$$x = e^t \cos t, \quad y = e^t \sin t, \quad z = 1$$

Soln: Let $\bar{r}$ be the position vector at any point (x, y, z) in a space.

$$\therefore \bar{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$= (e^t \cos t)\hat{i} + (e^t \sin t)\hat{j} + \hat{k} \quad \text{---(1)}$$

$$\therefore \text{Tangent} \cdot \frac{d\bar{r}}{dt} = \frac{d}{dt}\left[(e^t \cos t)\,\hat{i} + (e^t \sin t)\,\hat{j} + \hat{k}\right]$$

$$= (e^t \cos t - e^t \sin t)\hat{i} + (e^t \sin t + e^t \cos t)\,\hat{j} + 0\hat{k}$$

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$\therefore$ magnitude of tangent vector

$$\left|\frac{d\bar{r}}{dt}\right| = \sqrt{(e^t\cos t - e^t\sin t)^2 + (e^t\sin t + e^t\cos t)^2 + \cdots}$$

$$= \sqrt{e^{2t}(\cos t - \sin t)^2 + e^{2t}(\sin t + \cos t)^2}$$

$$= e^t \sqrt{\cos^2 t + 2\sin t\cos t + \sin^2 t + \sin^2 t + 2\sin t\cos t + \cos^2 t}$$

$$= e^t \sqrt{2\cos^2 t + 2\sin^2 t} = e^t \cdot \sqrt{2}$$

Hence

Unit tangent vector

$$\hat{t} = \frac{\frac{d\bar{r}}{dt}}{\left|\frac{d\bar{r}}{dt}\right|} = \frac{1}{e^t\sqrt{2}}\left[(e^t\cos t - e^t\sin t)\hat{i} + (e^t\sin t + e^t\cos t)\hat{j}\right]$$

$$\hat{t} = \frac{1}{\sqrt{2}}\left[(\cos t - \sin t)\hat{i} + (\sin t + \cos t)\hat{j}\right]$$

Now, accln ($\bar{a}$) = Rate of change of velocity ($\bar{v} = \frac{d\bar{r}}{dt}$)

$$= \frac{d^2\bar{r}}{dt^2} = \frac{d}{dt}\left(\frac{d\bar{r}}{dt}\right)$$

$$= \frac{d}{dt}\left[(e^t\cos t - e^t\sin t)\hat{i} + (e^t\sin t + e^t\cos t)\hat{j}\right]$$

$$\boxed{\bar{a} = -2e^t\sin t\,\hat{i} + 2e^t\cos t\,\hat{j}}$$

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∴ Tangential component of accln. ($\bar{a}$) = $\bar{a} \cdot \hat{t}$

$$= \left(-2e^t \sin t\, i + 2e^t \cos t\, j\right) \cdot \frac{1}{\sqrt{2}} \left[(\cos t - \sin t)\, i + (\cos t + \sin t)\, j\right]$$

$$a_t = \frac{1}{\sqrt{2}} \left[-2e^t \sin t\,(\cos t - \sin t) + 2e^t \cos t \cdot (\cos t + \sin t)\right]$$

$$= \frac{1}{\sqrt{2}} (+2e^t) \cdot \left[-\sin t \cos t + \sin^2 t + \cos^2 t + \cos t \sin t\right]$$

$$\boxed{a_t = \sqrt{2}\, e^t}$$

Normal component of accln ($\bar{a}$) = $|(\hat{t} \times \bar{a})|$

Here

$$\hat{t} \times \bar{a} = \begin{vmatrix} i & j & k \\ \frac{1}{\sqrt{2}}(\cos t - \sin t) & \frac{1}{\sqrt{2}}(\cos t + \sin t) & 0 \\ -2e^t \sin t & 2e^t \cos t & 0 \end{vmatrix}$$

$$= \frac{1}{\sqrt{2}} \cdot 2e^t \begin{vmatrix} i & j & k \\ \cos t - \sin t & \cos t + \sin t & 0 \\ -\sin t & \cos t & 0 \end{vmatrix}$$

$$= \sqrt{2} e^t \cdot \left[i(0-0) - j(0-0) + k(\cos^2 t - \sin t \cos t + \sin t \cos t + \sin^2 t)\right]$$

$$= \sqrt{2} e^t (0\hat{i} + 0\hat{j} + \hat{k})$$

$$= 0i + 0j + \sqrt{2} e^t \hat{k}$$

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$$\therefore a_n = |(\hat{t} \times \bar{a})|$$

$$= \sqrt{(0)^2 + (0)^2 + (\sqrt{2}e^t)^2}$$

$$= \sqrt{0 + 0 + 2e^{2t}} = \sqrt{2}\, e^t$$

$$\therefore \boxed{a_n = \sqrt{2}\, e^t}$$

H.W

① If $\bar{a}$ and $\bar{b}$ are constant vectors and ω is any scaler and $\bar{r} = \bar{a}\sin\omega t + \bar{b}\cos\omega t$ show that

(i) $\frac{d^2\bar{r}}{dt^2} + \omega^2 \bar{r} = 0$ (ii) $\bar{r} \times \frac{d\bar{r}}{dt} = -\omega \bar{a} \times \bar{b}$

② A partical moves along the curve

$x = 3t^2$, $y = t^2 - 2t$, $z = t^3$

find it's velocity and accln. at time $t = 1$ in the direction of vector $\hat{i} + \hat{j} - \hat{k}$

Hint:

velocity in direction of vector $i+j-k$ $= \frac{d\bar{r}}{dt} \cdot \hat{n}$

accn in direction of vector $i+j-k$ $= \frac{d^2\bar{r}}{dt^2} \cdot \hat{n}$

where $\hat{n} = \frac{\hat{i}+\hat{j}-\hat{k}}{\sqrt{1^2+1^2+1^2}} = \frac{1}{\sqrt{3}}(\hat{i}+\hat{j}-\hat{k})$.

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Vector differential operator

The vector differential operator Del (or nabla) is denoted and defined as

$$\nabla = i\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}$$

★ Gradient :⇒

Let $\phi(x,y,z)$ is scalar point function which is differentiable at each point (x,y,z), then gradient of ϕ denoted and define as -

$$\text{grad.}\,\phi = \nabla\phi$$

$$= \left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right)\phi(x,y,z)$$

$$= \frac{\partial\phi}{\partial x}\,\hat{i} + \frac{\partial\phi}{\partial y}\,\hat{j} + \frac{\partial\phi}{\partial z}\,k$$

Note: The component of $\nabla\phi$ in the direction of vector $\bar{a}$ is given by $\nabla\phi \cdot \hat{n}$ and it is called as directional derivative of ϕ in the direction of vector $\bar{a}$ where $\hat{n} = \frac{\bar{a}}{|\bar{a}|}$

Physically, this is the rate of change of ϕ at pt (x,y,z) in the direction of $\bar{a}$.

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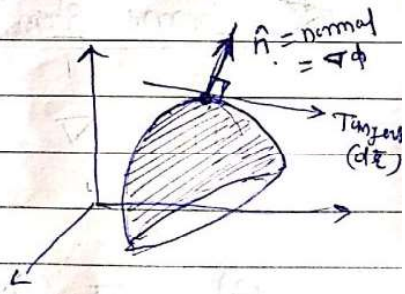
$\therefore$ **Directional derivative** $= \nabla\phi \cdot \hat{a} = \nabla\phi \cdot \frac{\bar{a}}{|a|}$

Geometrical meaning of $\nabla\phi$

Let $\phi(x,y,z) = c$ represent a family of surfaces for distinct value of const. c. Such surfaces for which the value of function is const is called as level surface. (Flat surface)

$$\therefore \boxed{d\phi = 0} \quad \longleftarrow (1)$$

Let $\bar{r} = x\hat{i} + y\hat{j} + z\hat{k}$ be the position vector of any point (x,y,z) on surface ϕ.



$\therefore d\bar{r} = dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}$

Now

By defn of total differentiation

$$d\phi = \frac{\partial\phi}{\partial x}\cdot dx + \frac{\partial\phi}{\partial y}\cdot dy + \frac{\partial\phi}{\partial z}\cdot dz$$

$$= \left(\frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\cdot\hat{k}\right) \cdot \left(dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}\right)$$

$$= \nabla\phi \cdot d\bar{r} \quad \longleftarrow (2)$$

$\therefore$ from (1) & (2)

$$0 = \nabla\phi \cdot d\bar{r}.$$

$\Rightarrow$ vector $\nabla\phi$ and $d\bar{r}$ are $\perp$er to each other.

Thus ($\nabla\phi$ represent a normal vector to the surface ϕ.)

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Since vector $d\bar{r}$ is in the direction of tangent vector to the given surface.

$\Rightarrow$ vector $\nabla\phi$ is $\perp^{er}$ to the tangent to the surface ϕ

and Hence $\nabla\phi$ is in the ~~normal~~ direction of normal to the surface.

Thus geometrically $\nabla\phi$ represent a vector normal to the given surface $\phi(x,y,z) = c$.



★ Some Standard Results

① $\nabla(\phi \pm \psi) = \nabla\phi \pm \nabla\psi$

② $\nabla(\phi \cdot \psi) = \phi \cdot (\nabla\psi) + (\nabla\phi) \cdot \psi$

③ $\nabla \circ f(u) = \hat{i}\frac{\partial f(u)}{\partial x} + \hat{j}\frac{\partial f(u)}{\partial y} + \hat{k}\frac{\partial f(u)}{\partial z}$

$= \hat{i} f'(u)\frac{\partial u}{\partial x} + \hat{j} f'(u)\frac{\partial u}{\partial y} + \hat{k} f'(u)\frac{\partial u}{\partial z}$

$= f'(u) \cdot \left[\hat{i}\frac{\partial u}{\partial x} + \hat{j}\frac{\partial u}{\partial y} + \hat{k}\frac{\partial u}{\partial z}\right]$

$= f'(u) \circ \nabla u$

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Ex :⇒ Find the angle betn the surfaces

$x^2+y^2+z^2 = 9$ and $x^2+y^2-z^2 = 1$ at pt. $(2,-1,2)$

Soln: Given surfaces

$$\phi_1(x,y,z) = x^2+y^2+z^2-9 \quad \text{---(1)}$$

$$\phi_2(x,y,z) = x^2+y^2-z-1 \quad \text{---(2)}$$

∴ Normal vector to surface ϕ_1

$$\therefore \nabla\phi_1 = i\frac{\partial\phi_1}{\partial x} + j\frac{\partial\phi_1}{\partial y} + k\frac{\partial\phi_1}{\partial z}$$

$$= \hat{i}(2x) + \hat{j}(2y) + \hat{k}(2z)$$

$$\nabla\phi_1 = 2[x\hat{i}+y\hat{j}+z\hat{k}] \quad \text{---(3)}$$

and $|\nabla\phi_1| = \sqrt{\quad}$

at pt $(2,-1,2)$ ⋯ $\nabla\phi_1 = 4\hat{i} - 2\hat{j} + 4\hat{k}$

Hence $|\nabla\phi_1| = \sqrt{(4)^2+(-2)^2+(4)^2} = \sqrt{36} = 6$

Now

Normal vector to surface (ϕ_2)

i.e $\nabla\phi_2 = \hat{i}\frac{\partial\phi_2}{\partial x} + \hat{j}\frac{\partial\phi_2}{\partial y} + \hat{k}\frac{\partial\phi_2}{\partial z}$

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$$\nabla\phi_2 = 2x\hat{i} + 2y\hat{j} - 2\hat{k}$$

at pt. $(2,-1,2)$,

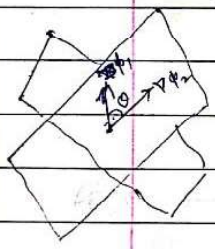
$$\nabla\phi_2 = 4\hat{i} - 2\hat{j} - 4\hat{k} \qquad \text{---------- (3)}$$

and $$|\nabla\phi_2| = \sqrt{(4)^2 + (2)^2 + (4)^2} = \sqrt{36} = 6$$

Now

angle betn two surface at pt $(2,-1,2)$. is nothing but angle betn its normal vectors $\nabla\phi_1$ & $\nabla\phi_2$ at pt $(2,-1,2)$.

$$\therefore \text{angle } \theta = \cos^{-1}\left\{\frac{\nabla\phi_1 \cdot \nabla\phi_2}{|\nabla\phi_1| \cdot |\nabla\phi_2|}\right\}$$

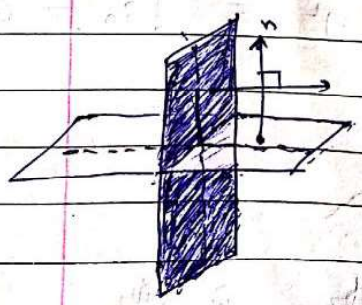


$$= \cos^{-1}\left\{\frac{(4\hat{i} - 2\hat{j} + 4\hat{k}) \cdot (4\hat{i} - 2\hat{j} - 4\hat{k})}{6 \cdot 6\sqrt{21}}\right\}$$

$$\theta = \cos^{-1}\left(\frac{8}{3\sqrt{21}}\right) = \cos^{-1}\left(\frac{16 + 4 - 16}{36}\right)$$

$$= \cos^{-1}\left(\frac{4}{36}\right)$$

$$\boxed{\theta = \cos^{-1}\left(\frac{1}{9}\right)}$$



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Ex: Find the directional derivative of surface $\phi = xy^2 + yz^3$ at pt $(2,-1,1)$ in the direction of the normal vector to the surface $x\log z - y^2 = -4$ at $(-1,2,1)$.

Sol^n: Let

$$\psi = x\log z - y^2 + 4 \quad \text{be the surface}$$

Then, its normal vector is given by

$$\nabla\psi = \left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right)\cdot\psi$$

$$= \left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right)\cdot(x\log z - y^2 + 4)$$

$$= i(\log x) + j(2y) + \hat{k}\left(\frac{x}{z}\right)$$

$$\therefore \nabla\psi = (\log z)\,\hat{i} + (2y)\hat{j} + \left(\frac{x}{z}\right)\hat{k}$$

at pt $(-1,2,1)$, $\nabla\psi = (\log 1)\,\hat{i} + 2(2)\hat{j} + \left(\frac{-1}{1}\right)\hat{k}$

$$\boxed{\nabla\psi = 0\,\hat{i} + 4\hat{j} - \hat{k}}$$

also. $|\nabla\psi| = \sqrt{0 + (4)^2 + (-1)^2} = \sqrt{17}$.

Now,

let $\phi = xy^2 + yz^3$ be the surface

$$\therefore \nabla\phi = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial z}\hat{k} + \frac{\partial\phi}{\partial y}\hat{k}$$

$$= \frac{\partial}{\partial x}(xy^2 + yz^3)\,\hat{i} + \frac{\partial}{\partial y}(xy^2 + yz^3)\hat{j} + \frac{\partial}{\partial z}(xy^2 + yz^3)\hat{k}$$

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$$\nabla\phi = (y^2)\hat{i} + (2xy+z^3)\hat{j} + (3yz^2)\hat{k}$$

at pt: $(2,-1,1)$

$$\nabla\phi = (-1)^2\hat{i} + (2(2)(-1)+(1)^3)\hat{j} + 3(-1)\cdot(1)^2\hat{k}$$

$$\Rightarrow \nabla\phi = 0\cdot\hat{i} - 3\hat{j} - 3\hat{k}$$

By defn. of Directional derivative

The directional derivative of a surface $\phi = xy^2 + yz^3$ in the direction of normal vector to surface ψ is

$$D.D = \nabla\phi \circ \frac{\nabla\psi}{|\nabla\psi|}$$

$$= (i-3j-3k)\cdot\frac{(0i-4j-k)}{\sqrt{17}}$$

$$= \frac{1}{\sqrt{17}}(0+12+3)$$

$$\therefore \boxed{D.D = \frac{15}{\sqrt{17}}}$$

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Ex: Find the directional derivative of $\phi = 4e^{2x-y+z}$ at the point $(1,1,-1)$ in the direction towards the point $(-3,5,6)$.

Soln: Given surface is

$$\phi(x,y,z) = 4e^{2x-y+z} \quad \text{---(1)}$$

$$\therefore \nabla\phi = i\frac{\partial\phi}{\partial x} + j\frac{\partial\phi}{\partial y} + k\frac{\partial\phi}{\partial z}$$

$$= i\left(8e^{2x-y+z}\right) + j\left(-4e^{2x-y+z}\right) + \hat{k}\left(4e^{2x-y+z}\right)$$

at pt. $(1,1,-1)$

$$\nabla\phi = i\left[8e^{(2-1-1)}\right] + \hat{j}\left[-4e^{2-1-1}\right] + \hat{k}\left[4e^{2-1-1}\right.$$

$$\boxed{\nabla\phi = 8\hat{i} - 4\hat{j} + 4\hat{k}}$$

Now a vector from pt A(1,1,-1) to pt B(-3,5,6) is given by

$$\overline{AB} = (-3-1)\hat{i} + (5-1)\hat{j} + (6-(-1))\hat{k}$$

$$= -4\hat{i} + 4\hat{j} + 7\hat{k}$$

$$\therefore |\overline{AB}| = \sqrt{(-4)^2+(4)^2+(7)^2} = \sqrt{81} = 9$$

∴ Direction derivative of surface ϕ in direction of vector $\overline{AB}$ is

$$D.D. = \nabla\phi \cdot \frac{\overline{AB}}{|AB|} = (8\hat{i}-4\hat{j}+4\hat{k}) \cdot \frac{(-4\hat{i}+4\hat{j}+7\hat{k})}{9}$$

$$= \frac{(-32-16+28)}{9} = \frac{-20}{9}$$

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Ex :⇒ Find the directional derivative of $\phi = 6x^2y + 24y^2z - 8z^2x$ at pt (1, 1, 1) in the direction parallel to the line $\frac{x-1}{2} = \frac{y-3}{-2} = \frac{z}{1}$ Hence find its maximum magnitude.

Soln: Given surface is

$$\phi = 6x^2y + 24y^2z - 8z^2x \quad \text{---(1)}$$

$$\therefore \nabla\phi = i\frac{\partial\phi}{\partial x} + j\frac{\partial\phi}{\partial y} + k\frac{\partial\phi}{\partial z}$$

$$= i\left[12xy + 0 - 8z^2\right] + j\left[6x^2 + 48yz\right] + k\left[24y^2 - 16zx\right]$$

at pt (1,1,1)

$$\nabla\phi = i(12-8) + j(6+48) + k(24-16)$$

$$\Rightarrow \boxed{\nabla\phi = 4\hat{i} + 54\hat{j} + 8\hat{k}}$$

Now, Given line

$$\frac{x-1}{2} = \frac{y-3}{-2} = \frac{z}{1}$$

It's Directional ratios are 2, −2, 1

∴ Direction vector of line is $2\bar{i} - 2\bar{j} + k = \bar{a}$ (say)

$$\Rightarrow |\bar{a}| = \sqrt{2^2 + 2^2 + 1^2} = \sqrt{9} = 3$$

Hence

$$D.D = \nabla\phi \cdot \frac{\bar{a}}{|\bar{a}|} = (4\bar{i} + 54\bar{j} + 8k) \cdot \frac{(2\bar{i} - 2\bar{j} + k)}{3}$$

$$= \frac{108}{3} = 36$$

Note

A vector parallel to the line l (whose Direction ratios are a, b, c) is given by $a\hat{i} + b\hat{j} + c\hat{k}$

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jno

magnitude of
Maximum value of ϕ (D.D.) $= |\nabla \phi|$

$= \sqrt{4^2 + (54)^2 + (8)^2} = \sqrt{2996}$

The greatest rate of change of ϕ i.e maximum of directional derivative is takes place in direction of normal vector $\nabla \phi$. and it's magnitude is $|\nabla \phi|$

Ex :⇒ Find the directional derivative of $\phi = e^{2x} \cdot \cos yz$ at the origin. In the direction of the tangent to the curve $x = a\sin t$, $y = a\cos t$, $z = at$ at $t = \pi/4$
also find magnitude of maximum value of directional derivative.

Soln :⇒ let

$$\phi = e^{2x} \cdot \cos(yz) \quad \text{---(1)}$$

$$\therefore \nabla \phi = \hat{i}\frac{\partial \phi}{\partial x} + \hat{j}\frac{\partial \phi}{\partial y} + \hat{k}\frac{\partial \phi}{\partial z}$$

$$= \hat{i}\,2[e^{2x}\cos(yz)] + \hat{j}\left(-z e^{2x}\sin(yz)\right) + \hat{k}\left(-y e^{2x}\sin(yz)\right)$$

at origin $(0,0,0)$

$$\boxed{\nabla \phi = 2\hat{i} + 0\hat{j} + 0\hat{k}}$$

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Now: let $\bar{r}$ be the position vector of any pt. (x, y, z) given by

$$\bar{r} = x\,i + y\,j + z\,k$$

$$\Rightarrow \bar{r} = (a\sin t)\,\hat{i} + (a\cos t)\,\hat{j} + (at)\,\hat{k}$$

$\therefore$ Tangent vector.

$$\frac{d\bar{r}}{dt} = \frac{d}{dt}\left[a\sin t\,\hat{i} + a\cos t\,\hat{j} + at\,\hat{k}\right]$$

$$= a\cos t\,\hat{i} - a\sin t\,\hat{j} + a\,\hat{k}$$

at $t = \pi/4$

$$\frac{d\bar{r}}{dt} = a\cos\left(\frac{\pi}{4}\right)\hat{i} - a\sin\left(\frac{\pi}{4}\right)\hat{j} + a\,\hat{k}$$

$$\frac{d\bar{r}}{dt} = \frac{a}{\sqrt{2}}\hat{i} - \frac{a}{\sqrt{2}}\hat{j} + a$$

$$\therefore \left|\frac{d\bar{r}}{dt}\right| = \sqrt{\left(\frac{a}{\sqrt{2}}\right)^2 + \left(\frac{-a}{\sqrt{2}}\right)^2 + (a)^2} = \sqrt{\frac{4a^2}{2}} = a\sqrt{2}$$

Now: Directional derivative of surface ϕ in the direction of tangent is vector.

$$D.D. = \nabla\phi \cdot \frac{(d\bar{r}/dt)}{|d\bar{r}/dt|}$$

$$= \frac{1}{a\sqrt{2}}(2i + 0j + 0k) \cdot \left(a/\sqrt{2}\,i - a/\sqrt{2}\,j + a\right)$$

$$= \frac{a}{a\sqrt{2}}\left[\left(2 \cdot \frac{a}{\sqrt{2}}\right) + 0 + 0\right]$$

$$\boxed{D.D = 1\,a}$$

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The maximum of D.D. is in the direction of vector normal

$\nabla\phi = 2\hat{i} + 0\hat{j} + 0\hat{k}$

and the magnitude of its maximum value $= |\nabla\phi|$

$= \sqrt{(2)^2 + 0 + 0} = \sqrt{4} = 2$

xoox

(HW) (1) Find the directional derivative of $xy^2 + yz^3$ at the point $(2,-1,1)$ along the tangent to the curve

$x = at\sin t$, $y = a\cos t$, $z = at$, at $t = \pi/4$

Ans: $D.D = \dfrac{16 + \pi - 12\sqrt{2}}{\sqrt{64 + 8\pi + \pi^2}}$

(2) Find the rate of change (Directional Derivative) of $\phi = xyz$ in the direction of normal to the surface $x^2y + y^2x + yz^2 = 3$ at pt $(1,1,1)$

$D.D. = \dfrac{9}{\sqrt{29}}$

(3) Find the directional derivative of $\phi(x,y,z) = x^2 - 2y^2 + 4z^2$ at the point $(1,1,-1)$ in the direction of $2\hat{i} + \hat{j} - \hat{k}$, In what Direction will the directional derivative be maximum, and what is it's magnitude?

$D.D = 8/\sqrt{6}$

D.D is maximum in the direction of vector

$\nabla\phi = 2i - 4j - 8k$

and magnitude $= |\nabla\phi| = \sqrt{84} = 2\sqrt{21}$

(4) Find the Directional derivative of $1/r$ in the direction of position vector $\bar{r}$ at any pt (x,y,z).

Hint: Here $\phi = 1/r = \dfrac{1}{\sqrt{x^2+y^2+z^2}}$

$\dfrac{d}{dx}\dfrac{1}{x} = \dfrac{d}{dx}(x^{-1}) = -\dfrac{1}{x^2}$

$= -\dfrac{1}{2}\dfrac{1}{x^{3/2}}$

$\nabla\phi = -\dfrac{x}{2\sqrt{x^2+y^2+z^2}}\dfrac{\partial\phi}{\partial x}i + \dfrac{y}{y}j + \dfrac{z}{z}k$

$= -\dfrac{x}{(x^2+y^2+z^2)^{3/2}}i + \dfrac{y}{(x^2+y^2+z^2)^{3/2}}j - \dfrac{z}{(x^2+y^2+z^2)^{3/2}}k$

$\nabla\phi = -\dfrac{1}{(x^2+y^2+z^2)^{3/2}}[xi + yj + zk]$

Hint $= \bar{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$\therefore r = \sqrt{x^2+y^2+z^2}$

$\therefore D.D = \nabla\left(\dfrac{1}{r}\right) \cdot \left(\dfrac{\bar{r}}{|\bar{r}|}\right)$

$= -1/r^2 = \dfrac{-1}{x^2+y^2+z^2}$

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★ Divergence ⇒

Let $\vec{V}(x,y,z) = V_1\hat{i} + V_2\hat{j} + V_3\hat{k}$ be a vector point function which is differentiable at each point (x,y,z) in a certain region of space. (i.e. V define a vector field)

Then the divergence of $\vec{V}$ is denoted and defined as

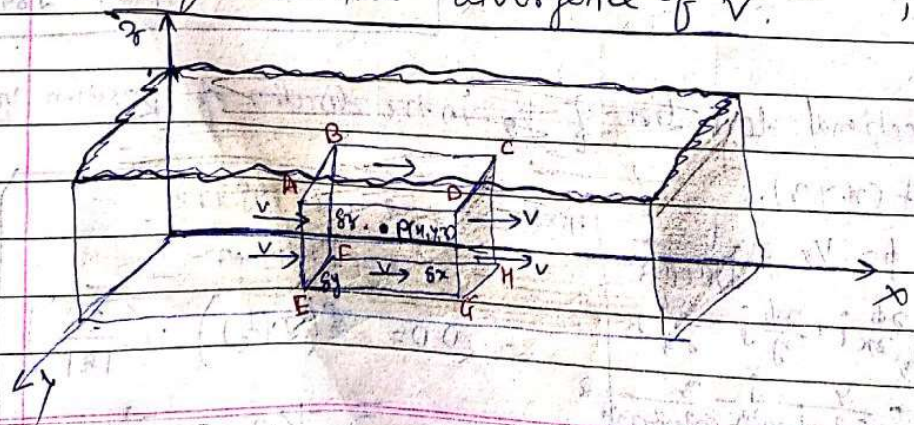
$$\nabla \cdot \vec{V} = \left(\frac{\partial}{\partial x}i + \frac{\partial}{\partial y}j + \frac{\partial}{\partial z}k\right) \cdot \left(V_1 i + V_2 j + V_3 k\right)$$

$$\boxed{\nabla \cdot \vec{V} = \frac{\partial V_1}{\partial x} + \frac{\partial V_2}{\partial y} + \frac{\partial V_3}{\partial z}}$$

Note that $\nabla \cdot \vec{V} \neq \vec{V} \cdot \nabla$

Physical interpretation of divergence of $\vec{V}$

"If $\vec{V}(x,y,z) = V_1\hat{i} + V_2\hat{j} + V_3\hat{k}$ be the velocity of moving fluid at any point $P(x,y,z)$, passing through a parallelopied whose dimensions are x, y & z. then total gain of a fluid per unit volume per unit time is equal to the divergence of $\vec{V}$." i.e. $(\nabla \cdot \vec{V})$



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Here: x component of velocity V is $= V_1$

$\therefore$ x component of V at centre of face BCFH $= \left(V_1 - \frac{1}{2}\frac{\partial V_1}{\partial x}\cdot \delta x\right)$ approx.

also

x component of V at centre of face EGAD $= \left(V_1 + \frac{1}{2}\frac{\partial V_1}{\partial x}\delta x\right)$ approx.

Thus

Volume of a fluid crossing BCFH per unit time $= \left(V_1 - \frac{1}{2}\frac{\partial V_1}{\partial x}\delta x\right)\cdot \delta y\, \delta z$ — (1)

Volume of a fluid crossing EGAD per unit time $= \left(V_1 + \frac{1}{2}\frac{\partial V_1}{\partial x}\delta x\right)\delta y \cdot \delta z$ — (2)

$\therefore$ Total volume of fluid per unit time in x-direction = (2) − (1)

$$= \frac{\partial V_1}{\partial x}\delta x \delta y \delta z$$

Similarly

Total volume of fluid per unit time in y-direction $= \frac{\partial V_2}{\partial y}\cdot \delta x \delta y \delta z$

&

Total volume of fluid per unit time in z-direction $= \frac{\partial V_3}{\partial z}\cdot \delta x \delta y \delta z$

Then Total gain of fluid per unit time per unit volume $= \frac{\left(\frac{\partial V_1}{\partial x} + \frac{\partial V_2}{\partial y} + \frac{\partial V_3}{\partial z}\right)\cdot \delta x \delta y \delta z}{\delta x \delta y \delta z}$

$$= \frac{\partial V_1}{\partial x} + \frac{\partial V_2}{\partial y} + \frac{\partial V_3}{\partial z}$$

$$= \nabla \cdot \vec{V}.$$

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★ Irrotational field

A vector field $\vec{V}$ is said to be irrotational iff

$\text{Curl } \vec{V} = \nabla \times \vec{V} = 0$ at all points of the field.

$\Longleftrightarrow$

Note.

If a vector field $\vec{F} = \nabla \phi$ where ϕ is scalar point function.

Then

$$\text{Curl } \vec{F} = \nabla \times (\nabla \phi)$$

$$= 0$$

Then $\vec{F}$ is called as conservative vector field.

i.e. If $\text{Curl } \vec{F} = \nabla \times \vec{F} = 0$ then we can find the scalar point function $\phi(x, y, z)$ using the relation

$$\boxed{\vec{F} = \nabla \phi}$$

where $\vec{F}$ is conservative vector field.

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✱ Properties of Divergence :

① $\nabla \cdot (\bar{f} \pm \bar{g}) = \nabla \cdot \bar{f} \pm \nabla \cdot \bar{g}$

② $\nabla \cdot (\phi \cdot \bar{f}) = (\nabla \cdot \phi) \cdot \bar{f} + \phi (\nabla \cdot \bar{f})$ where ϕ is Scalar function.

③ $\nabla \cdot \vec{F} = 0$ if $\vec{F}$ is const. vector (solenoidal function).

④ $\nabla \cdot (\bar{f} \times \bar{g}) = \bar{g} \cdot (\nabla \times \vec{f}) - \bar{f} \cdot (\nabla \times \bar{g})$

⑤ $\nabla \cdot (\nabla \cdot \phi) \equiv \nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2}$

where $\nabla^2 = \left(\frac{\partial^2}{\partial x^2} \hat{i} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)$ is known as Laplacian operator.

✱ Properties of curl

① $\nabla \times (\bar{f} + \bar{g}) = (\nabla \times \bar{f}) + (\nabla \times \bar{g})$

② $\nabla \times (\phi \cdot \bar{f}) = (\nabla \cdot \phi) \times \bar{f} + \phi \cdot (\nabla \times \bar{f})$

③ $\nabla \times (\nabla \cdot \phi) = 0$

④ $\nabla \times (\bar{f} \times \bar{g}) = (\bar{g} \cdot \nabla) \cdot \bar{f} - (\nabla \cdot \bar{f}) \cdot \bar{g} + \bar{f} \cdot (\nabla \cdot \bar{g}) - \bar{g} (\nabla \cdot \bar{f})$

⑤ $\nabla \times (\nabla \times \bar{f}) = \nabla \cdot (\nabla \cdot f) - \nabla^2 f$

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Ex :⇒ Find the directional derivative of the divergence of vector $\vec{F} = xy\,\hat{i} + xy^2\,\hat{j} + z^2\,\hat{k}$ at pt. $(2,1,2)$ in the direction of outer normal to the sphere $x^2+y^2+z^2=9$.

Soln: let

$$\vec{F} = xy\,\hat{i} + xy^2\,\hat{j} + z^2\,\hat{k}$$

$\therefore$ div. $\vec{F} = \nabla \cdot \vec{F}$

$$= \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right) \cdot \left(xy\,\hat{i} + xy^2\,\hat{j} + z^2 k\right)$$

$$= \frac{\partial}{\partial x}(xy) + \frac{\partial}{\partial y}(xy^2) + \frac{\partial}{\partial z}(z^2)$$

$$\therefore \boxed{\text{div}\cdot\vec{F} = y + 2xy + 2z} = \phi(x,y,z) \text{ say.}$$

Now. Normal vector to the surface $\phi = \text{div}\cdot\vec{F} = y+2xy+2z$ is given by

$$\nabla\phi = \left(\frac{\partial\phi}{\partial x} i + \frac{\partial\phi}{\partial y} \cdot j + \frac{\partial\phi}{\partial z} k\right)$$

$$= 2y\,i + (1+2x)\,j + 2k$$

at pt $(2,1,2)$

$$\nabla\phi = 2i + 5j + 2k$$

Now, consider a given sphere is

$$\psi = x^2+y^2+z^2-9$$

$\therefore$ its normal vector is given by

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$$\nabla\psi = \frac{\partial\psi}{\partial x}\hat{i} + \frac{\partial\psi}{\partial y}\hat{j} + \frac{\partial\psi}{\partial z}\hat{k}$$

$$= 2x\hat{i} + 2y\hat{j} + 2z\hat{k}$$

at pt. (2,1,2)

$$\nabla\psi = 4\hat{i} + 2\hat{j} + 4\hat{k}$$

also

$$|\nabla\psi| = \sqrt{4^2+2^2+4^2} = \sqrt{36} = 6$$

Now the directional derivative of div. $\vec{F} = \phi$ in the direction of normal vector to surface ψ (i.e. $\nabla\psi$) at pt(2,1,2) is

$$D.D = \nabla\phi \cdot \frac{\nabla\psi}{|\nabla\psi|}$$

$$= (2\hat{i}+5\hat{j}+2\hat{k}) \cdot \frac{(4\hat{i}+2\hat{j}+4\hat{k})}{6}$$

$$= \frac{8+10+8}{8} = \frac{26}{6} = \frac{13}{3}.$$

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Ex: Show that $\vec{F} = (y^2\cos x + z^3)\hat{i} + (2y\sin x - 4)\hat{j} + (3xz^2+2)\hat{k}$ is a conservative vector field. and Hence find a scalar potential function ϕ such that $\vec{F} = \nabla\phi$

Also find the work done in moving particle from pt $(0, 1, -1)$ to $(\pi/2, -1, 2)$.

Soln: let

$$\vec{F} = (y^2\cos x + z^3)\hat{i} + (2y\sin x - 4)\hat{j} + (3xz^2+2)\hat{k} \quad \text{---(1)}$$

$$\therefore \text{curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y^2\cos x + z^3 & 2y\sin x - 4 & 3xz^2+2 \end{vmatrix}$$

$$= \hat{i}(0-0) - \hat{j}(3z^2 - 3z^2) + \hat{k}(2y\cos x - 2y\cos x)$$

$$= 0\hat{i} + 0\hat{j} + 0\hat{k} = 0$$

since $\text{curl } \vec{F} = 0 \Rightarrow \vec{F}$ is conservative vector field.

(or $\vec{F}$ is irrotational vector field.)

To find ϕ

let us consider a conservative vector field $\vec{F}$ is given by

$$\vec{F} = \nabla\phi$$

$$\Rightarrow (y^2\cos x + z^3)\hat{i} + (2y\sin x - 4)\hat{j} + (3xz^2+2)\hat{k}$$

$$= \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}$$

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equating coefficient of i, j and k we get

$$\frac{\partial \phi}{\partial x} = y^2\cos x + z^3 \quad \text{——(1)}$$

$$\frac{\partial \phi}{\partial y} = 2y\sin x - 4 \quad \text{——(2)}$$

$$\frac{\partial \phi}{\partial z} = 3xz^2 + 2 \quad \text{——(3)}$$

Note : By defn of total Diff.

$$d\phi = \frac{\partial \phi}{\partial x}dx + \frac{\partial \phi}{\partial y}dy + \frac{\partial \phi}{\partial z}dz$$

$$\therefore \phi = \int_{y,z\ \text{const}} \frac{\partial \phi}{\partial x}\cdot dx + \int_{x,z\ \text{const}} \frac{\partial \phi}{\partial y}dy + \int_{xy\ \text{const}} \frac{\partial \phi}{\partial z}dz$$

$$= (y^2\sin x + z^3 x) + (y^2\sin x) - 4y + xz^3 + 2z$$

Cancell the terms ~~it is~~ if repeated term.

on integrating w.r.to. x, y, z (partially), & respectively we get

$$\phi(x,y,z) = y^2\sin x + xz^3 + \phi_1(y,z) \quad \text{——(4)}$$

$$\phi(x,y,z) = y^2\sin x - 4y + \phi_2(x,z) \quad \text{——(5)}$$

$$\phi(x,y,z) = xz^3 + 2z + \phi_3(x,y) \quad \text{——(6)}$$

Compairing (4) (5) & (6), we get

$\phi_1(y,z)$ = terms of ϕ not ~~containing~~ containing $x = 2z - 4y$

Similarly

$\phi_2(x,z)$ = Term of ϕ not containing $y = xz^3 + 2z$

$\phi_3(x,y)$ = Terms of ϕ not containing $z = y^2\sin x - 4y$

$\therefore$ From (3) $\boxed{\phi(x,y,z) = y^2\sin x + xz^3 + 2z - 4y}$

is req'd scalar potential function.

To find work done by potential function ϕ from pt. $A(0,1,-1)$ to $B(\pi/2, -1, 2)$.

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$\therefore$ Work done $= \phi(B) - \phi(A)$

$= \phi(\pi/2, -1, 2) - \phi(1, 1, -1)$

$= \left[(-1)^2 \sin\pi/2 + \pi/2(2)^3 - 4(-1) + 2(2)\right] - \left[1^2 \sin 0 + 0(-1)^3 - 4(1) + 2(-1)\right]$

$= (1 + 4\pi + 4 + 4) - (0 + 0 - 4 - 2)$

Work done $= 4\pi + 5$



Ex :⇒ Find the value of n for which the vector field $r^n \cdot \vec{r}$ will be solenoidal. also test wheather the vector field is irrotational or not.

Soln: Let $\vec{r}$ be the position vector of a point (x, y, z).

i.e $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$\therefore r = |\vec{r}| = \sqrt{x^2+y^2+z^2} = (x^2+y^2+z^2)^{1/2}$

$r^2 = |r| \cdot |r| = x^2 + y^2 + z^2$

Thus $r^n = (x^2+y^2+z^2)^{n/2}$

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$\therefore \quad r^n \cdot \vec{r} = (x^2+y^2+z^2)^{n/2} \quad [\because xi + yj + zk]$

$= x(x^2+y^2+z^2)^{n/2} \cdot \hat{i} + y(x^2+y^2+z^2)^{n/2} \cdot \hat{j}$

$+ z(x^2+y^2+z^2)^{n/2} \hat{k}$ —— (1)

Now: $\frac{\partial}{\partial x}(r^n \vec{r}) = \frac{\partial}{\partial x}\left[(x) \cdot (x^2+y^2+z^2)^{n/2}\right]$

$= (1)(x^2+y^2+z^2)^{n/2} + \frac{n}{2} x \cdot (x^2+y^2+z^2)^{\frac{n}{2}-1} \cdot 2x$

$\Rightarrow \quad = (x^2+y^2+z^2)^{n/2} + nx^2 (x^2+y^2+z^2)^{\frac{n}{2}-1}$

Similarly $\frac{\partial}{\partial y}\left[y \cdot (x^2+y^2+z^2)^{n/2}\right] = (x^2+y^2+z^2)^{n/2} + ny^2 (x^2+y^2+z^2)^{\frac{n}{2}-1}$

$\frac{\partial}{\partial z}\left[z (x^2+y^2+z^2)^{n/2}\right] = (x^2+y^2+z^2)^{n/2} + nz^2 (x^2+y^2+z^2)^{\frac{n}{2}-1}$

Now $\text{grad.}(r^n \vec{r}) = \nabla \cdot (r^n \vec{r})$

$= \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right) \cdot \left(x(x^2+y^2+z^2)^{n/2} i + y(x^2+y^2+z^2)^{n/2} j + z(x^2+y^2+z^2)^{n/2} \cdot k\right]$

$= \frac{\partial}{\partial x}\left[x(x^2+y^2+z^2)^{n/2}\right] + \frac{\partial}{\partial y}\left[y(x^2+y^2+z^2)^{n/2}\right] + \frac{\partial}{\partial z}\left[z(x^2+y^2+z^2)^{n/2}\right]$

$= \left[(x^2+y^2+z^2)^{n/2} + nx^2(x^2+y^2+z^2)^{\frac{n}{2}-1}\right] + \left[(x^2+y^2+z^2) + ny^2(x^2+y^2+z^2)^{\frac{n}{2}-1}\right]$

$+ \left[(x^2+y^2+z^2) + nz^2(x^2+y^2+z^2)^{\frac{n}{2}-1}\right]$

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$$= 3(x^2+y^2+z^2)^{n/2} + \frac{n}{2}(x^2+y^2+z^2)\cdot(x^2+y^2+z^2)^{n/2-1}$$

$$= 3(x^2+y^2+z^2)^{n/2} + n(x^2+y^2+z^2)^{n/2}$$

$$\nabla\cdot(r^n\vec{r}) = (3+n)(x^2+y^2+z^2)^{n/2}$$

Since it is given that $r^n\vec{r}$ is solenoidal

$\therefore$ we have $\nabla\cdot(r^n\vec{r}) = 0$

ie $(3+n)(x^2+y^2+z^2)^{n/2} = 0$

$\Rightarrow 3+n = 0 \Rightarrow \boxed{n=-3}$

Now,

$$\text{Curl}(r^n\vec{r}) = \nabla\times(r^n\cdot\vec{r})$$

$$= \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x(x^2+y^2+z^2)^{n/2} & y(x^2+y^2+z^2)^{n/2} & z(x^2+y^2+z^2)^{n/2} \end{vmatrix}$$

$$= (x^2+y^2+z^2)^{n/2}\begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x & y & z \end{vmatrix}$$

$$= (x^2+y^2+z^2)^{n/2}\left[i(0-0) + j(0-0) + k(0-0)\right]$$

$\nabla\times(r^n\vec{r}) = 0$

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⇒ vector field $\vec{r}$ is an irrotational vector field

Ex :⇒ Find the const. a and b so that the surfaces

$$5x^2 - 2yz - gx = 0 \quad \& \quad ax^2y + bz^3 = 4$$

may cut orthogonally at the point $(1, -1, 2)$.

Soln :⇒ let given surfaces are

$$\phi_1 = 5x^2 - 2yz - gx \qquad \text{---(i)}$$

$$\phi_2 = ax^2y + bz^3 - 4 \qquad \text{---(ii)}$$

Normal vector to the surface ϕ_1 is given by.

$$\nabla \phi_1 = \frac{\partial \phi_1}{\partial x} i + \frac{\partial \phi_1}{\partial y} j + \frac{\partial \phi_1}{\partial z} k$$

$$= (10x - g)\, i + (-2z)\, j + (-2y)\, \hat{k}$$

∴ at pt $(1, -1, 2)$

$$\boxed{\nabla \phi_1 = i - 4j + 2k} \qquad \text{---(iii)}$$

Normal vector to the surface ϕ_2 is

$$\nabla \phi_2 = \frac{\partial \phi_2}{\partial x} i + \frac{\partial \phi_2}{\partial y} j + \frac{\partial \phi_2}{\partial z} k$$

$$= (2axy)\, \hat{i} + (ax^2)\, \hat{j} + (3bz^2)\, \hat{k}$$

at pt $(1, -1, 2)$

$$\boxed{\nabla \phi_2 = -2a\, \hat{i} + a\, \hat{j} + 12b\, \hat{k}} \qquad \text{---(iv)}$$

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Consider that, surface ϕ_1 and ϕ_2 cut orthogonally to each other at pt. $(1,-1,2)$.

$\Rightarrow$ Their normal vectors are perpendicular to each other

Thus we have

$$\nabla\phi_1 \cdot \nabla\phi_2 = 0$$

$$\Rightarrow (\hat{i} - 4\hat{j} + 2\hat{k}) \cdot (-2a\hat{i} + a\hat{j} + 12b\hat{k}) = 0$$

$$\Rightarrow -2a - 4a + 24b = 0$$

$$\Rightarrow -6a = -24b \quad \Rightarrow \boxed{a = 4b}$$

Since both surfaces meets each other at pt $(1,-1,2)$

$\therefore \phi_2$ at $(1,-1,2) \Rightarrow a(1)^2(-1) + b(2^3) = 4$

$$\Rightarrow -a + 8b = 4$$

~~$a(1)^2(-1) + b(2)^3 = 4$~~

$$\Rightarrow -4b + 8b = 4$$

$$\Rightarrow 4b = 4 \quad \Rightarrow \boxed{b = 1}$$

$$\therefore \boxed{a = 4}$$

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(H.W)

Ex: ① Find the value of a so that vector field

$\vec{A} = (axy - z^3)\hat{i} + (a-2)x^2\hat{j} + (1-a)xz^2\hat{k}$

is an irrotational.

Hence find the scalar potential function ϕ so that it satisfy

$\vec{A} = \nabla\phi$.

Ans $\boxed{a = 4}$

$\boxed{\phi = 2x^2y - z^3x}$

Ex ② Find a, b, c so that a vector field

$\vec{F} = (x + 2y + az)\hat{i} + (bx - 3y - z)\hat{j} + (4x + cy + 2z)\hat{k}$

is conservative. Hence find scalar potential function such that $\vec{F} = \nabla\phi$.

Ans $a = 4,\ b = 2,\ c = -1$

$\phi = \frac{x^2}{2} - \frac{3y^2}{2} + z^2 + 2xy + 4zx - z$

Ex: ③ A vector field $\vec{V} = (x^2 + xy^2)\hat{i} + (y^2 + x^2y)\hat{j}$

is an irrotational. Hence find the scalar potential function ϕ such that $\vec{V} = \nabla\phi$.

Ans $\phi = x^3/3 + y^3/3 + \left(\frac{x^2 - y^2}{2}\right)$

Ex ④ Show that $\vec{E} = \vec{r}/r^2$ is irrotational.

Hence find ϕ so that $\vec{E} = -\nabla\phi$.

Hint: $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$\therefore |\vec{r}| = r = \sqrt{x^2 + y^2 + z^2}$

Ans $\phi = -\frac{1}{2}\log(x^2 + y^2 + z^2)$

Ex ⑤ Find the angle bet^n two surfaces

$x^2 + y^2 + z^2 = 9$ & $x^2 + y^2 - z^2 = 1$ at $(2, -1, 2)$

Hint: angle bet^n surface ϕ & ψ is the angle bet^n their normals ie $\nabla\phi$ & $\nabla\phi$

$\therefore \theta = \cos^{-1}\left(\frac{\nabla\phi \cdot \nabla\psi}{|\nabla\phi|\,|\nabla\psi|}\right)$

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Ex. (6) Find the value of n for which the vector field $r^n \vec{r}$ will be solenoidal. Also test whether the vector field is irrotational or not

Hint: (A vector $\vec{V}$ is said to be solenoidal iff $\nabla \cdot \vec{V} = 0$)

Ex (7) Find the const a & b so that the surface

$5x^2 - 2yz - gx = 0$ & $ax^2y + bz^3 = 4$

may cut each other orthogonally at pt $(1, -1, 2)$.

Ex (8) Show that

$$\text{Curl}\left(\frac{\vec{a} \times \vec{r}}{r^n}\right) = \frac{2-n}{r^n} \cdot \vec{a} + \frac{n \cdot (\vec{a} \cdot \vec{r})}{r^{n+2}} \cdot \vec{r}$$

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